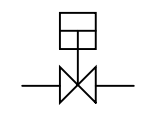
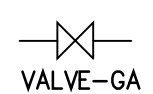


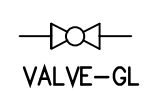
VALVE LEGEND



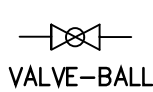
ON/OFF VALVE
W/ POSITION SWITCHES



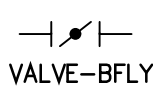
GATE VALVE
VALVE-GA



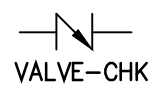
GLOBE VALVE
VALVE-GL



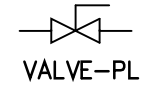
BALL VALVE
VALVE-BALL



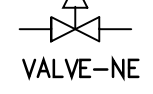
BUTTERFLY VALVE
VALVE-BFLY



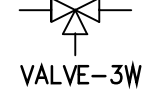
CHECK VALVE
VALVE-CHK



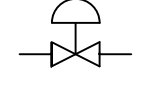
PLUG VALVE
VALVE-PL



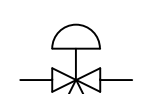
NEEDLE VALVE
VALVE-NE



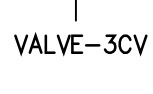
3 - WAY VALVE
VALVE-3W



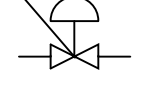
AUTOMATIC VALVE WITH
POSITIONER AND FEEDBACK



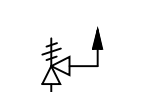
AUTOMATIC VALVE WITH
POSITIONER AND FEEDBACK



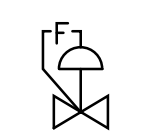
VALVE-3CV



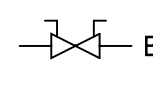
PRESSURE REDUCING VALVE
(PCV)



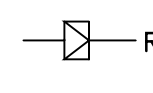
PRESSURE SAFETY VALVE



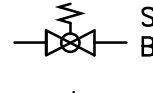
FAIL SAFE VALVE W/
FUSEABLE LINK



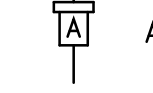
BALANCE VALVE



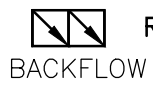
RUPTURE DISC



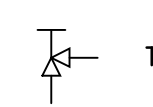
SPRING LOADED (DEADMAN)
BALL VALVE



AUTOMATIC AIR VENT

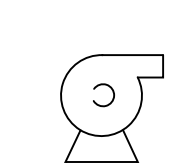


REDUCED PRESSURE BACKFLOW PREVENTER
BACKFLOW

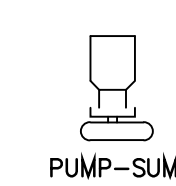


THROTTLING VALVE

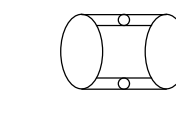
MISCELLANEOUS LEGEND



CENTRIFUGAL PUMP



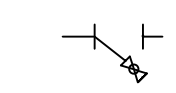
SUMP PUMP
PUMP-SUMP



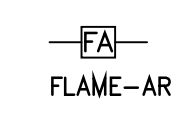
DIAPHRAM PUMP



HEAT EXCHANGER
HEAT EXC



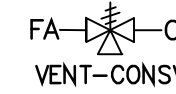
STRAINER WITH
BLOW DOWN VALVE



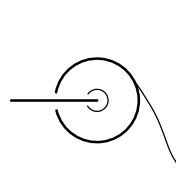
FLAME ARRESTOR
FLAME-AR



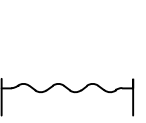
TRAP



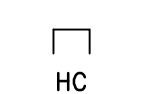
CONSERVATION VENT
FLAME ARRESTOR



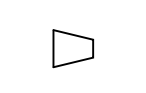
HOSE REEL



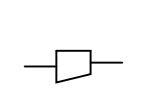
FLEXIBLE HOSE



HOSE CONNECTION
HC



REDUCER
RED



REDUCER-ECCENTRIC
RED-ECC



FLOW ORIFICE
FLOR



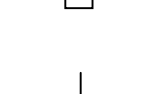
CAP
CAP



DRAIN



DRAIN



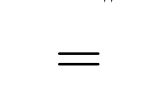
FILTER



SPRAY BALL



FLOW METER
FLOW TRANSMITTER



FLOW ARROW

PIPE FLANGE

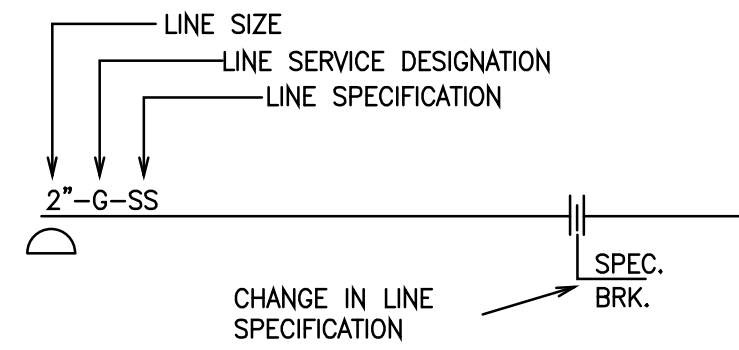
DEVICE NAMING LEGEND

VALVE - ON OFF	ONV-1234
PUMP	PMP-1234
AGITATOR	AGI±1234
CONVEYOR	CONV-1234
TURN HEADS	XXV-1234
SIDE GATE	XNV-1234
FAN	FAN-1234
TEMP CONTROL VALVE	TCV-1234
FLOW CONTROL VALVE	FCV-1234
LEVEL CONTROL VALVE	LCV-1234
OPENED LIMIT SWITCH	ZSA-1234
CLOSED LIMIT SWITCH	ZSB-1234
HIGH LEVEL SWITCH	LSH-1234
LOW LEVEL SWITCH	LSL-1234
LEVEL TRANSDUCER	LT-1234
FLOW TRANSDUCER	FT-1234
PRESSURE TRANSDUCER	PT-1234
TEMPERATURE TRANSMITTER	TT-1234
MOTOR OPERATED VALVE	MOV-1234
SPEED SWITCHES	SS-1234
JAM SWITCH	XS-1234
PRESSURE CONROL VALVE	PCV-1234
BALANCE VALVE	BAL-1234
RUPTURE DISC	RD-1234

PIPING MATERIAL SPECIFICATIONS

CS	CARBON STEEL
SS	STAINLESS STEEL PIPE
SST	STAINLESS STEEL OD TUBING
PP	PLASTIC, PVC/CPVC
CU	COPPER

LINE SPECIFICATION LEGEND



CONTINUATION

FROM

CL14-XXX-PI00X

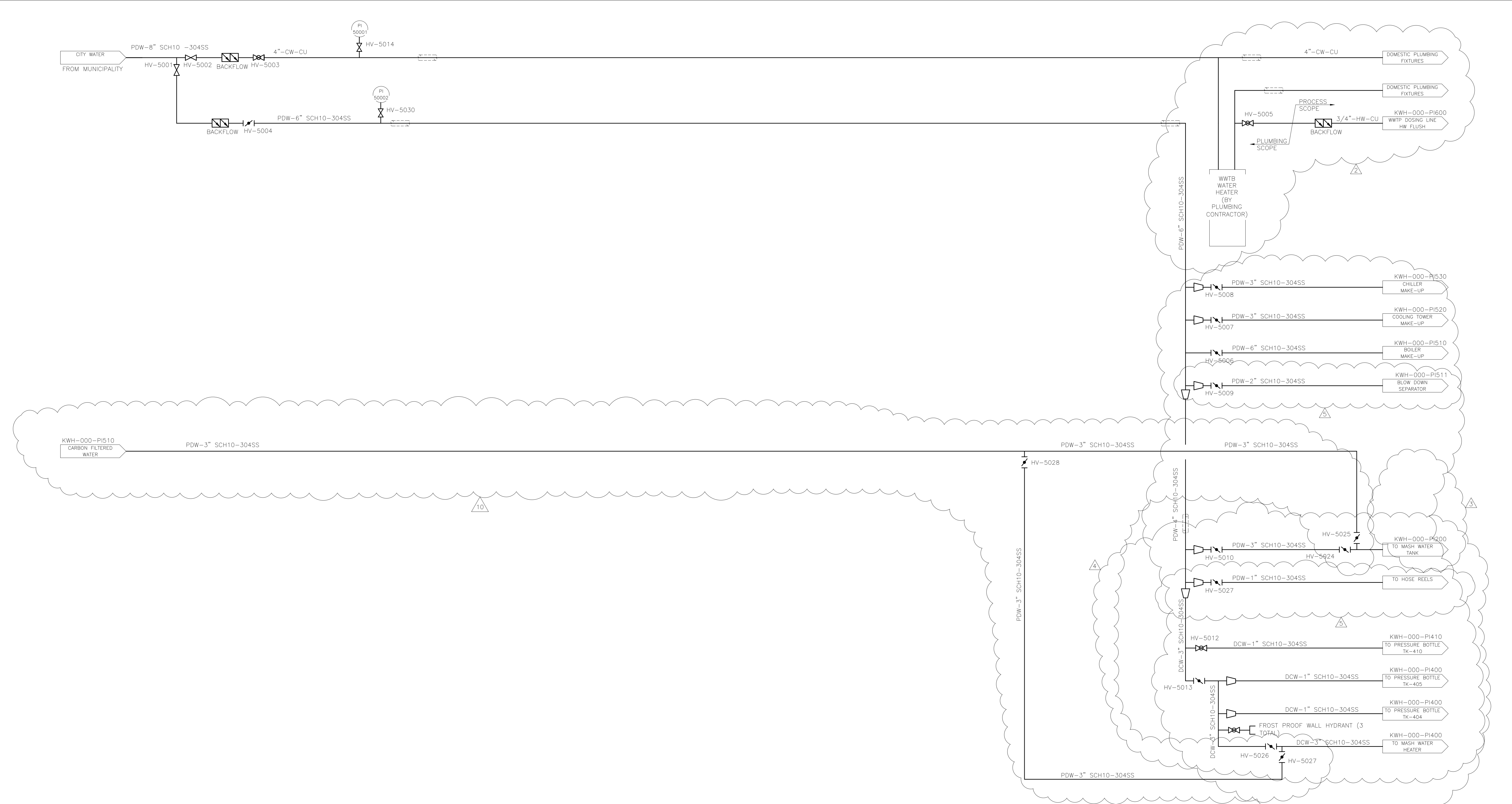
TO

CL14-XXX-PI00X

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NO.	REVISIONS	DATE	CHECKED APPROVED
WHISKEY HOUSE OF KENTUCKY			
DRAWING NO.	PI000	DRAWN BY ARH	DATE 7-22-22

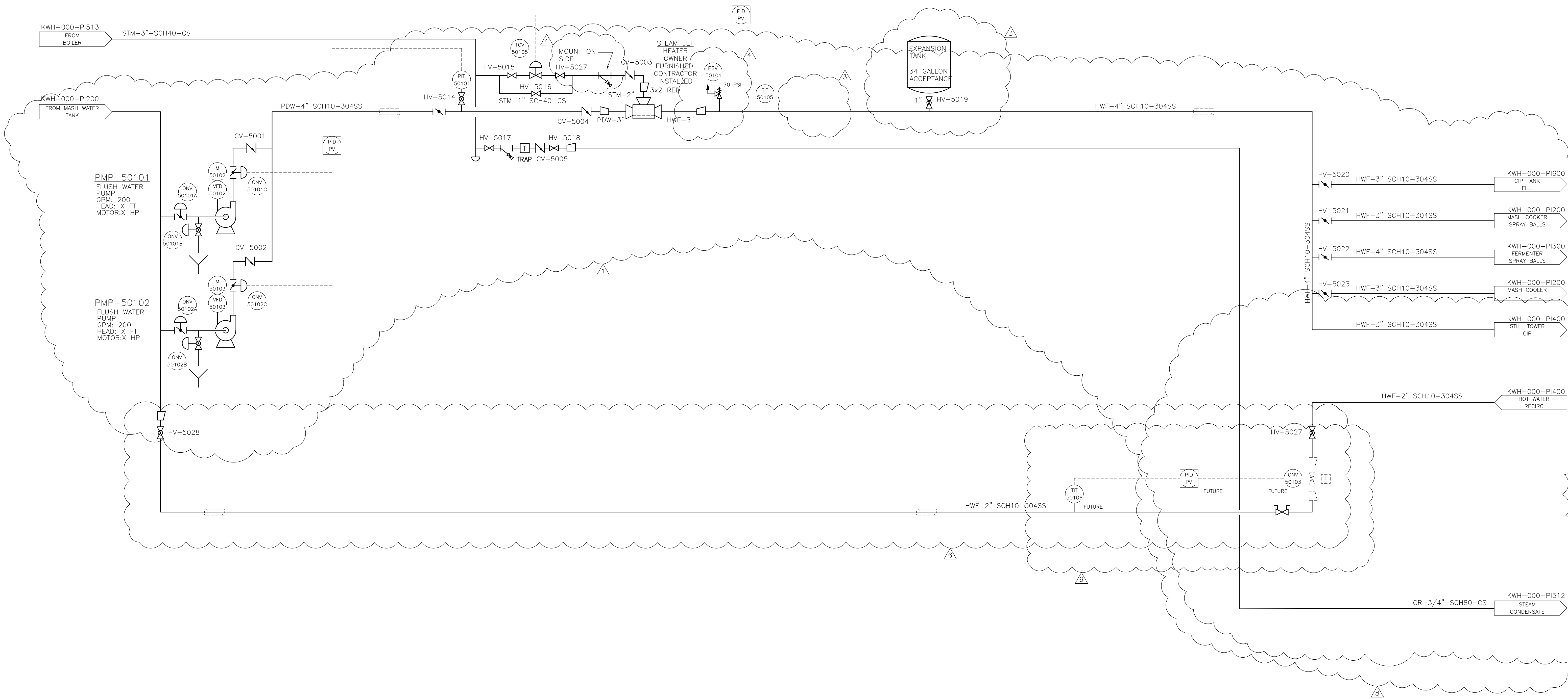


10	CARBON FILTER MASH WATER	4-29-24	
5	ROUTING UPDATES	7-10-23	
4	ROUTING UPDATES	6-23-23	
3	ROUTING UPDATES	5-2-23	
2	ROUTING UPDATES	3-31-23	
	PRICING	9-30-22	
	PRELIMINARY	9-5-22	
NO.	REVISIONS	DATE	CHECKED APPROVED

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WHISKEY HOUSE OF KENTUCKY PROCESS DOMESTIC WATER		
DRAWING NO.	KWH-000-PI500	DRAWN BY ARH
DATE	7-22-22	



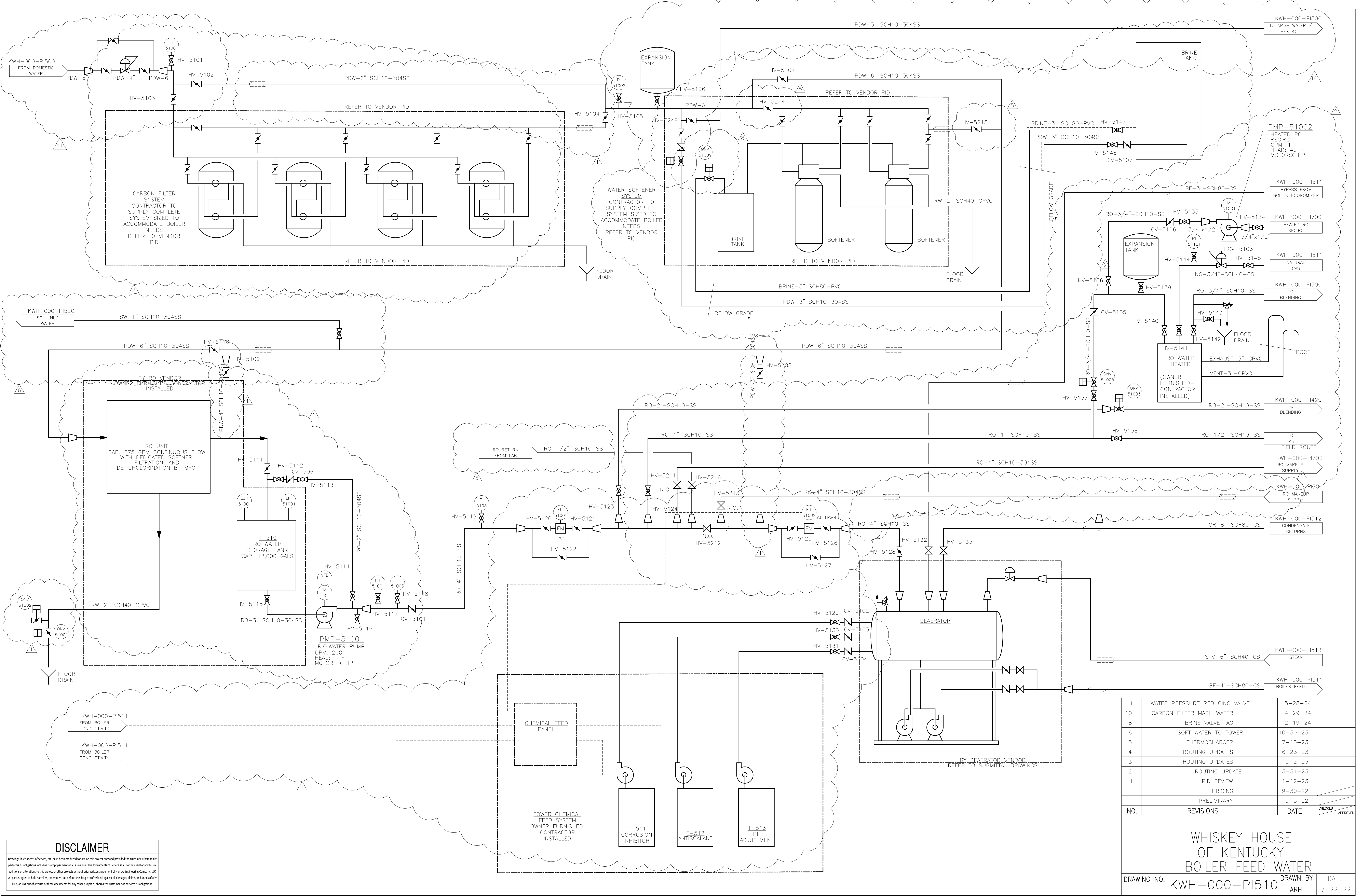
9	RECIRC LINE	4-15-24	
8	RECIRC LINE	2-19-24	
6	RECIRC LINE	10-30-23	
4	ROUTING UPDATES	6-23-23	
3	ROUTING UPDATES	5-2-23	
2	ROUTING UPDATE	3-31-23	
1	PID REVIEW	1-12-23	
	PRICING	9-30-22	
	PRELIMINARY	9-5-22	
NO.	REVISIONS	DATE	CHECKED APPROVED

WHISKEY HOUSE
OF KENTUCKY
HOT FLUSH WATER

DRAWING NO. KWH-000-PI501 DRAWN BY ARH DATE 7-22-22

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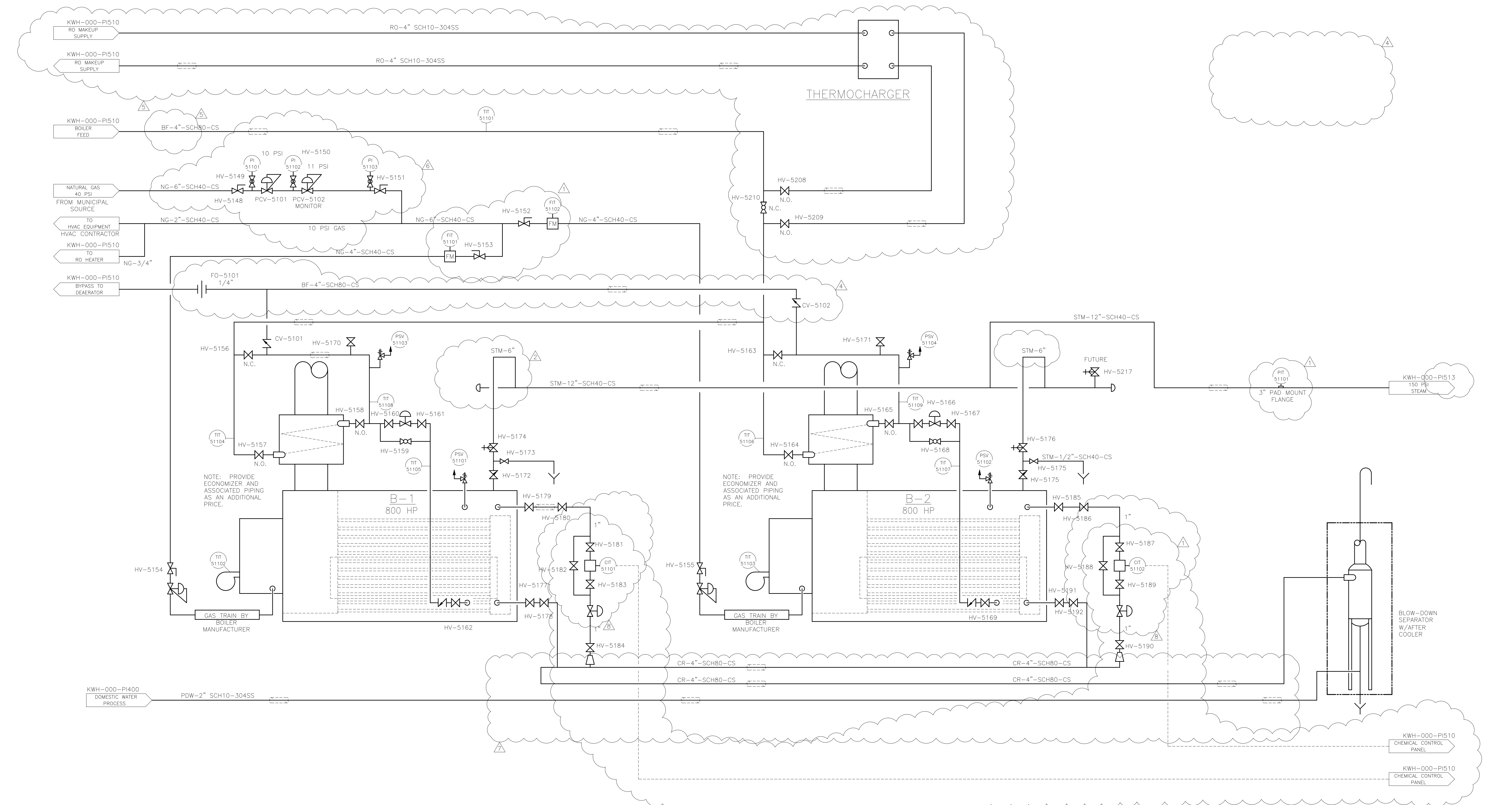
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11	WATER PRESSURE REDUCING VALVE	5-28-24	
10	CARBON FILTER MASH WATER	4-29-24	
8	BRINE VALVE TAG	2-19-24	
6	SOFT WATER TO TOWER	10-30-23	
5	THERMOCHARGER	7-10-23	
4	ROUTING UPDATES	6-23-23	
3	ROUTING UPDATES	5-2-23	
2	ROUTING UPDATE	3-31-23	
1	PID REVIEW	1-12-23	
	PRICING	9-30-22	
	PRELIMINARY	9-5-22	
NO.	REVISIONS	DATE	CHECKED APPROVED

WHISKEY HOUSE
OF KENTUCKY
BOILER FEED WATER

DRAWING NO. KWH-000-PI510
DRAWN BY ARH
DATE 7-22-22



DISCLAIMER

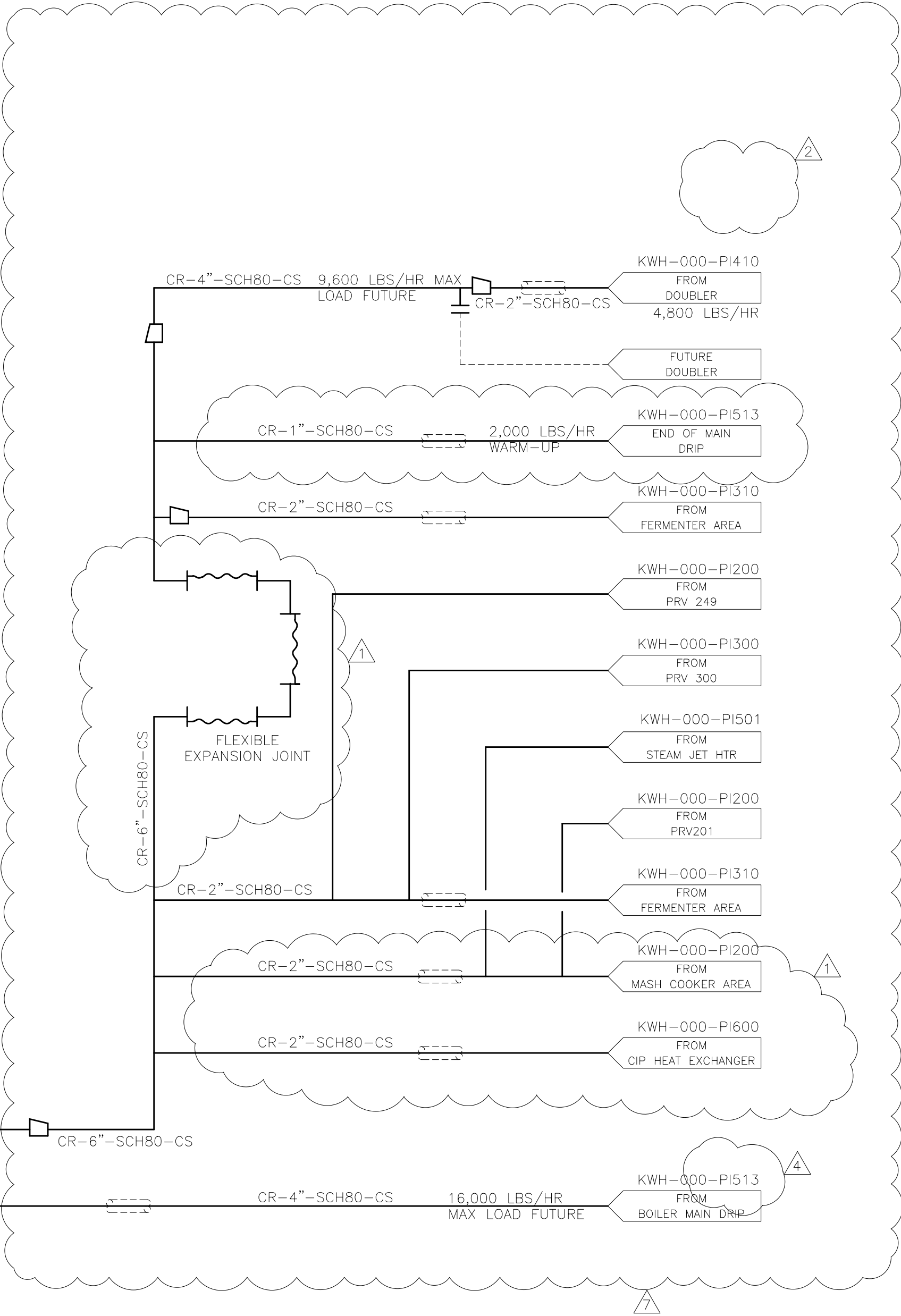
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7	ROUTING UPDATES	11-14-23	
6	ROTARY METER DELETED	8-28-23	
5	THERMOCHARGER	7-10-23	
4	ROUTING UPDATES	6-23-23	
2	ROUTING UPDATE	3-31-23	
1	PID REVIEW	1-12-23	
	PRICING	10-06-22	
	PRELIMINARY	9-5-22	
NO.	REVISIONS	DATE	CHECKED APPROVED

WHISKEY HOUSE OF KENTUCKY BOILER PIPING			
DRAWING NO.	KWH-000-PI511	DRAWN BY	DATE
		ARH	7-22-22

DISCLAIMER

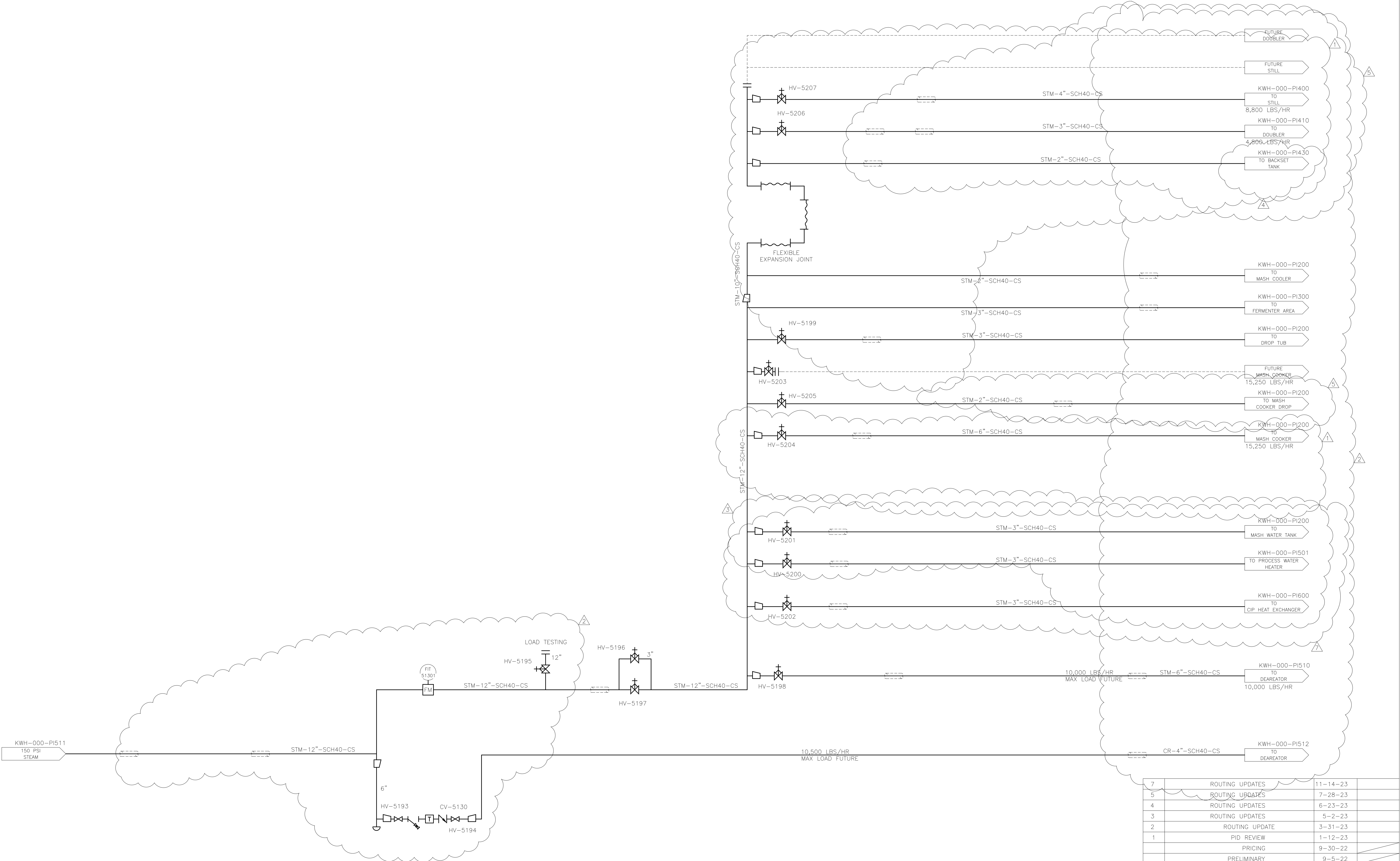
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7	ROUTING UPDATES	11-14-23	
4	ROUTING UPDATES	6-23-23	
3	ROUTING UPDATES	5-2-23	
2	ROUTING UPDATE	3-31-23	
1	PID REVIEW	1-12-23	
	PRICING	9-30-22	
	PRELIMINARY	9-5-22	
NO.	REVISIONS	DATE	CHECKED APPROVED
WHISKEY HOUSE OF KENTUCKY STEAM CONDENSATE			
DRAWING NO.	KWH-000-PI512	DRAWN BY ARH	DATE 7-22-22

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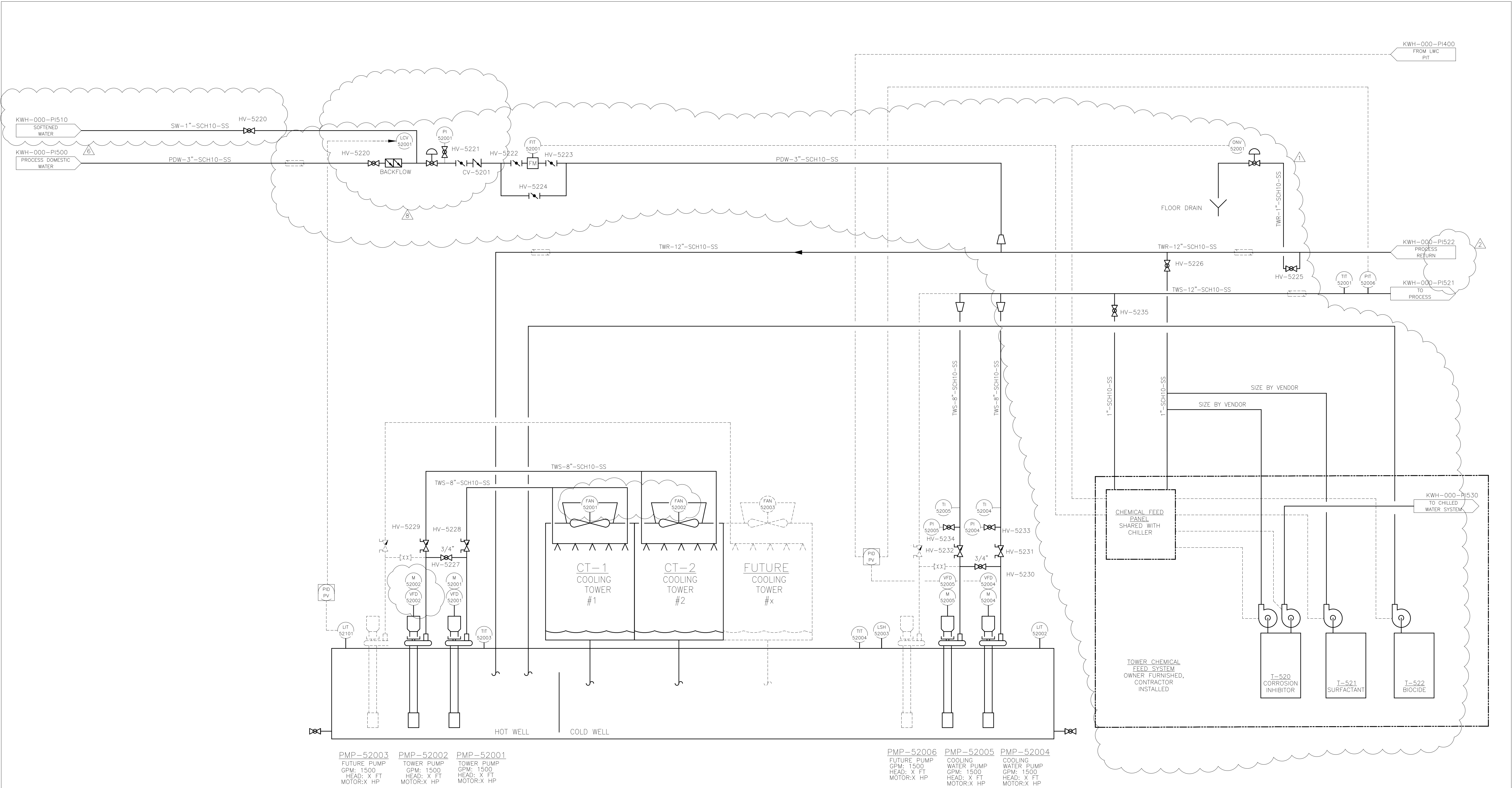
7	ROUTING UPDATES	11-14-23	
5	ROUTING UPDATES	7-28-23	
4	ROUTING UPDATES	6-23-23	
3	ROUTING UPDATES	5-2-23	
2	ROUTING UPDATE	3-31-23	
1	PID REVIEW	1-12-23	
	PRICING	9-30-22	
	PRELIMINARY	9-5-22	
NO.	REVISIONS	DATE	CHECKED APPROVED

WHISKEY HOUSE
OF KENTUCKY
STEAM SUPPLY

DRAWING NO. KWH-000-PI513

DRAWN BY ARH

DATE 7-22-22



PMP-52003
FUTURE PUMP
GPM: 1500
HEAD: X FT
MOTOR: X HP

PMP-52002
TOWER PUMP
GPM: 1500
HEAD: X FT
MOTOR: X HP

PMP-52001
TOWER PUMP
GPM: 1500
HEAD: X FT
MOTOR: X HP

PMP-52006
FUTURE PUMP
GPM: 1500
HEAD: X FT
MOTOR: X HP

PMP-52005
COOLING WATER PUMP
GPM: 1500
HEAD: X FT
MOTOR: X HP

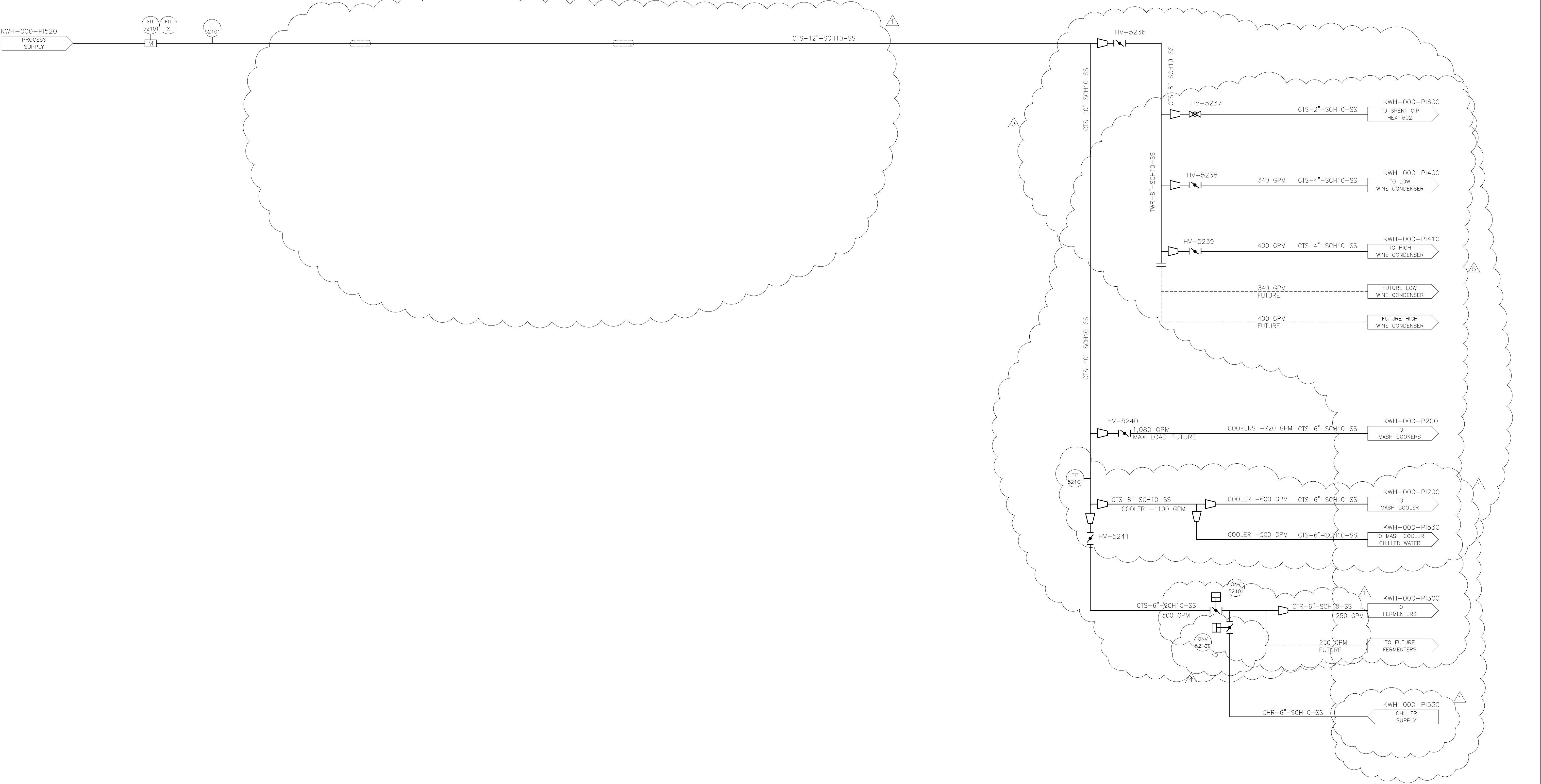
PMP-52004
COOLING WATER PUMP
GPM: 1500
HEAD: X FT
MOTOR: X HP

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8	SOFT WATER MAKE-UP	2-19-24	
6	SOFT WATER TO TOWER	10-30-23	
4	ROUTING UPDATES	6-23-23	
2	ROUTING UPDATE	3-31-23	
1	PID REVIEW	1-12-23	
	PRICING	9-30-22	
	PRELIMINARY	9-5-22	
NO.	REVISIONS	DATE	CHECKED APPROVED

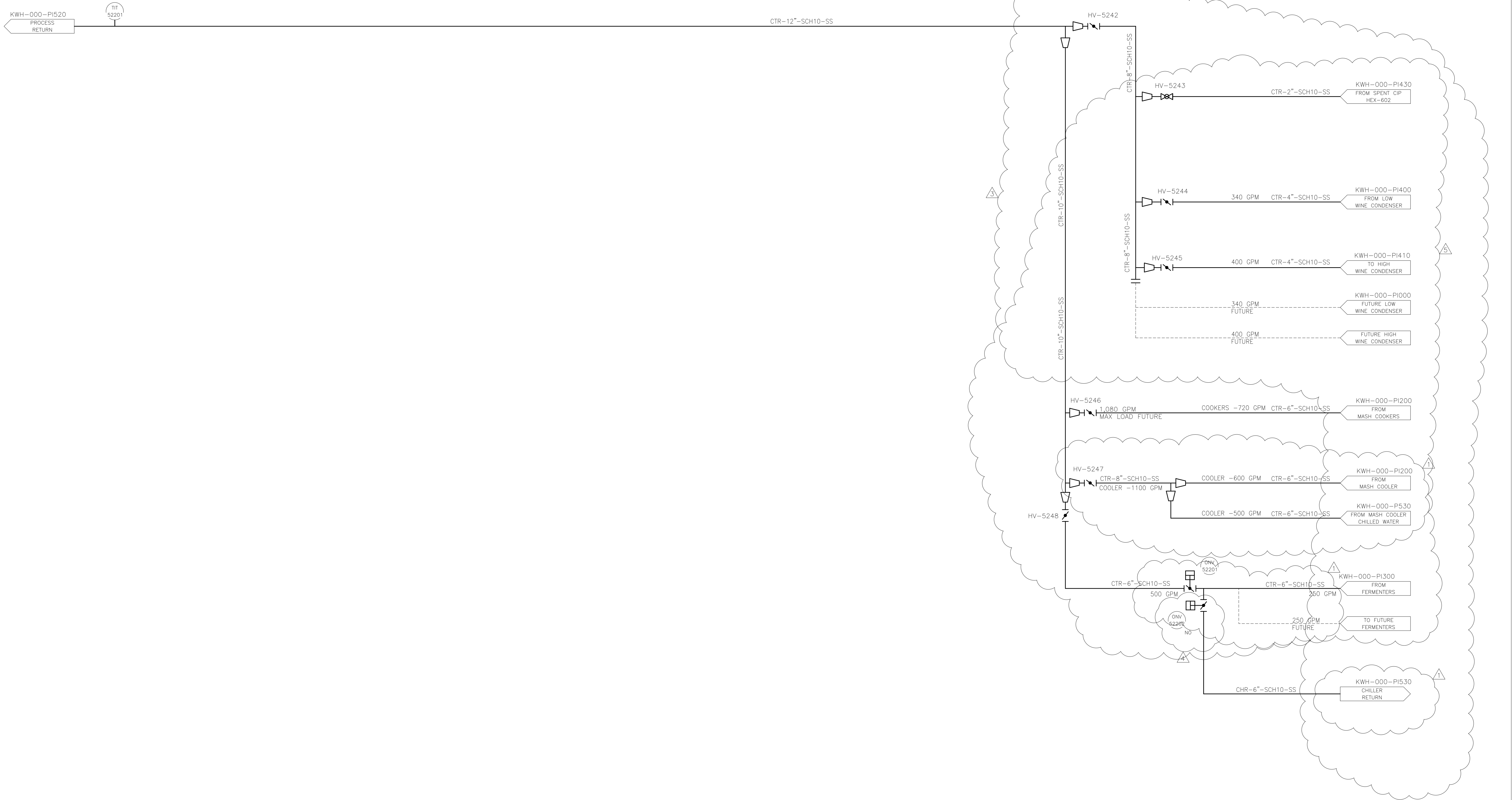
WHISKEY HOUSE OF KENTUCKY COOLING TOWER PIPING		
DRAWING NO.	KWH-000-PI520	DRAWN BY ARH
DATE	7-22-22	



5	ROUTING UPDATES	7-28-23	
4	ROUTING UPDATES	6-23-23	
3	ROUTING UPDATES	5-2-23	
2	ROUTING UPDATE	3-31-23	
1	PID REVIEW	1-12-23	
	PRICING	9-30-22	
	PRELIMINARY		
NO.	REVISIONS	DATE	CHECKED APPROVED

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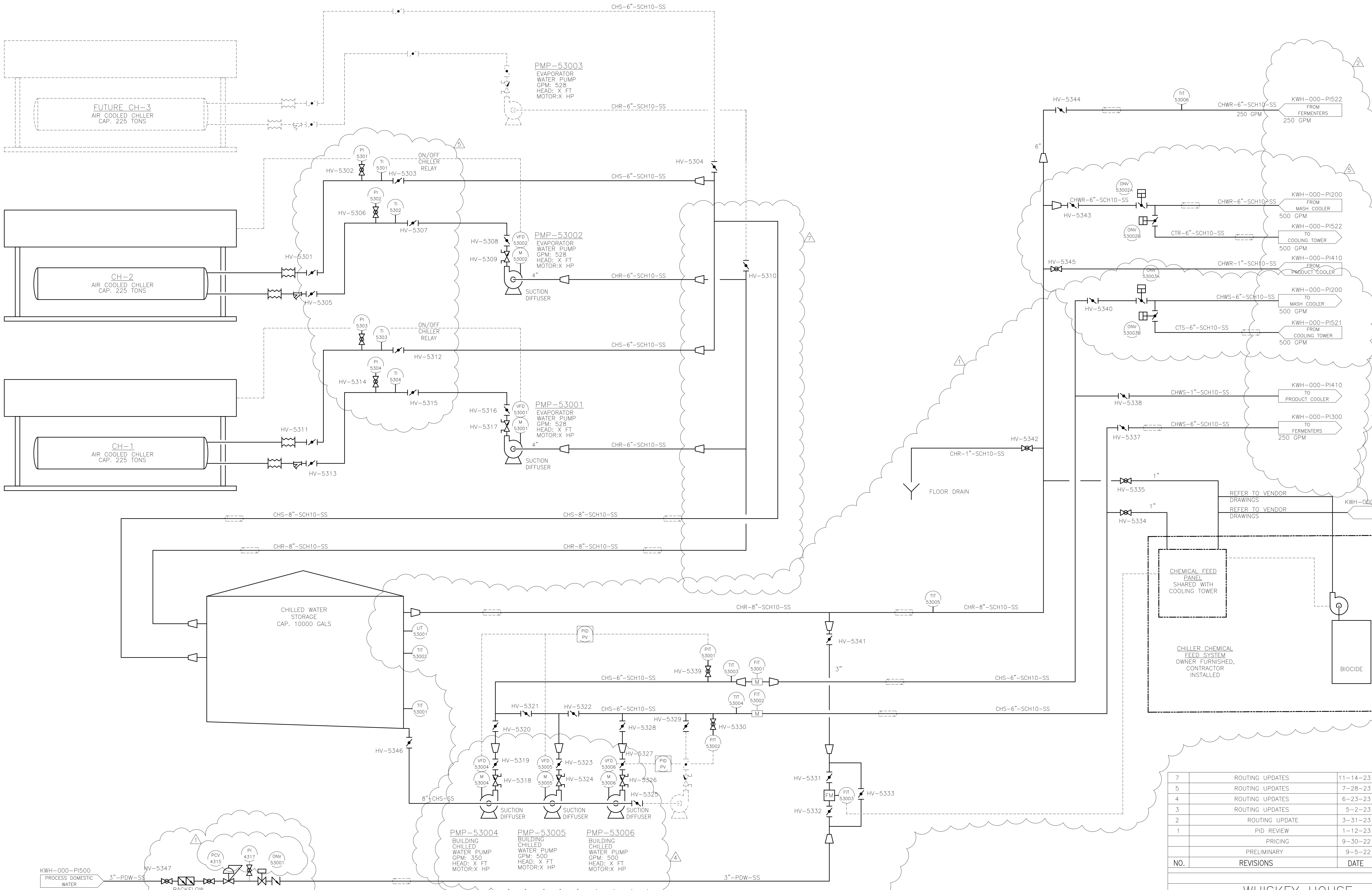
5	ROUTING UPDATES	7-28-23	
4	ROUTING UPDATES	6-23-23	
3	ROUTING UPDATES	5-2-23	
2	ROUTING UPDATE	3-31-23	
1	PID REVIEW	1-12-23	
	PRICING	9-30-22	
	PRELIMINARY	9-5-22	
NO.	REVISIONS	DATE	CHECKED APPROVED

WHISKEY HOUSE
OF KENTUCKY
COOLING TOWER RETURN

DRAWING NO. KWH-000-PI522DRAWN BY ARHDATE 7-22-22

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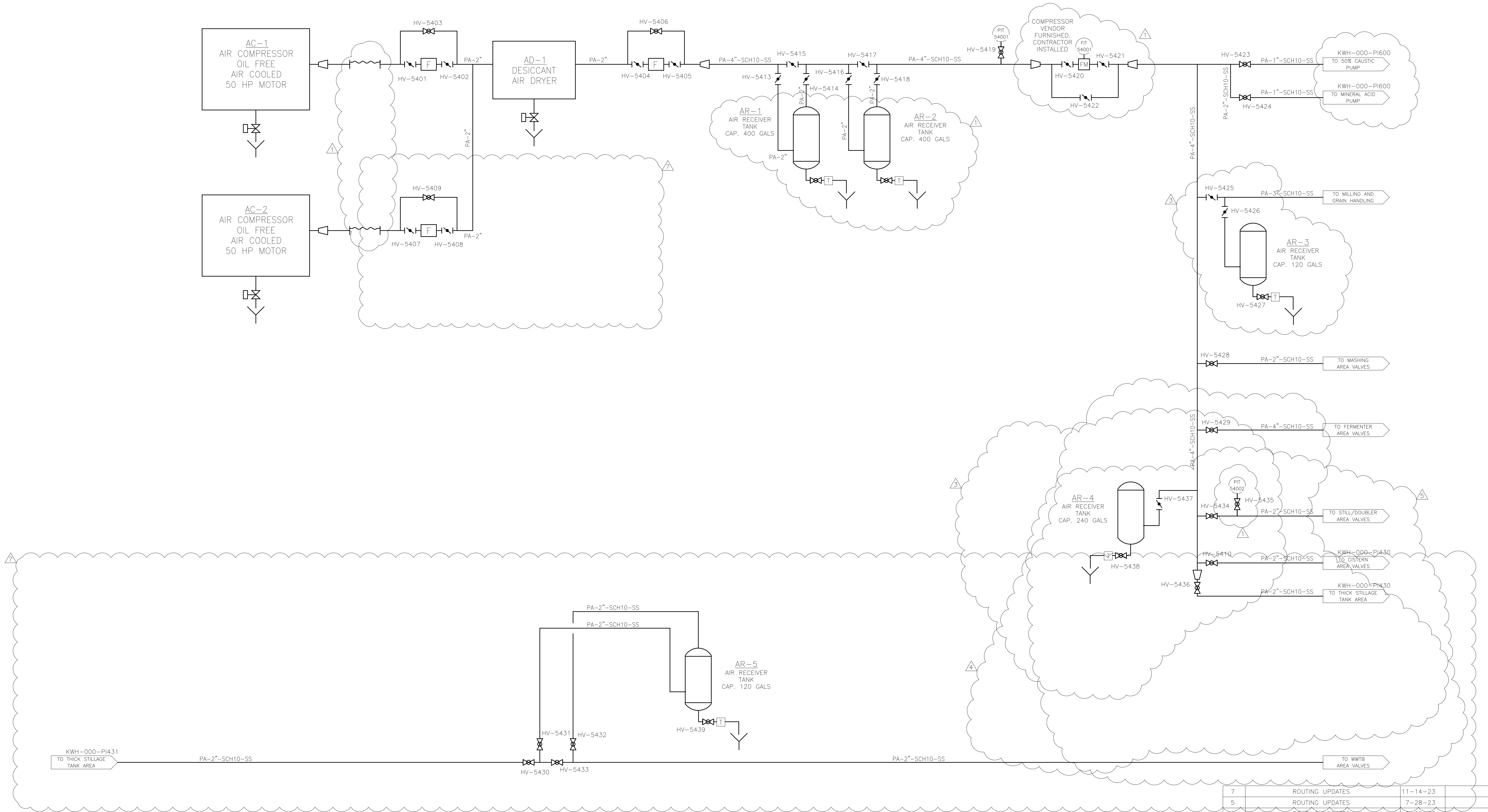
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7	ROUTING UPDATES	11-14-23	
5	ROUTING UPDATES	7-28-23	
4	ROUTING UPDATES	6-23-23	
3	ROUTING UPDATES	5-2-23	
2	ROUTING UPDATE	3-31-23	
1	PID REVIEW	1-12-23	
	PRICING	9-30-22	
	PRELIMINARY	9-5-22	
NO.	REVISIONS	DATE	CHECKED APPROVED

WHISKEY HOUSE
OF KENTUCKY
CHILLER PIPING

DRAWING NO. KWH-000-PI530 DRAWN BY ARH DATE 7-22-22



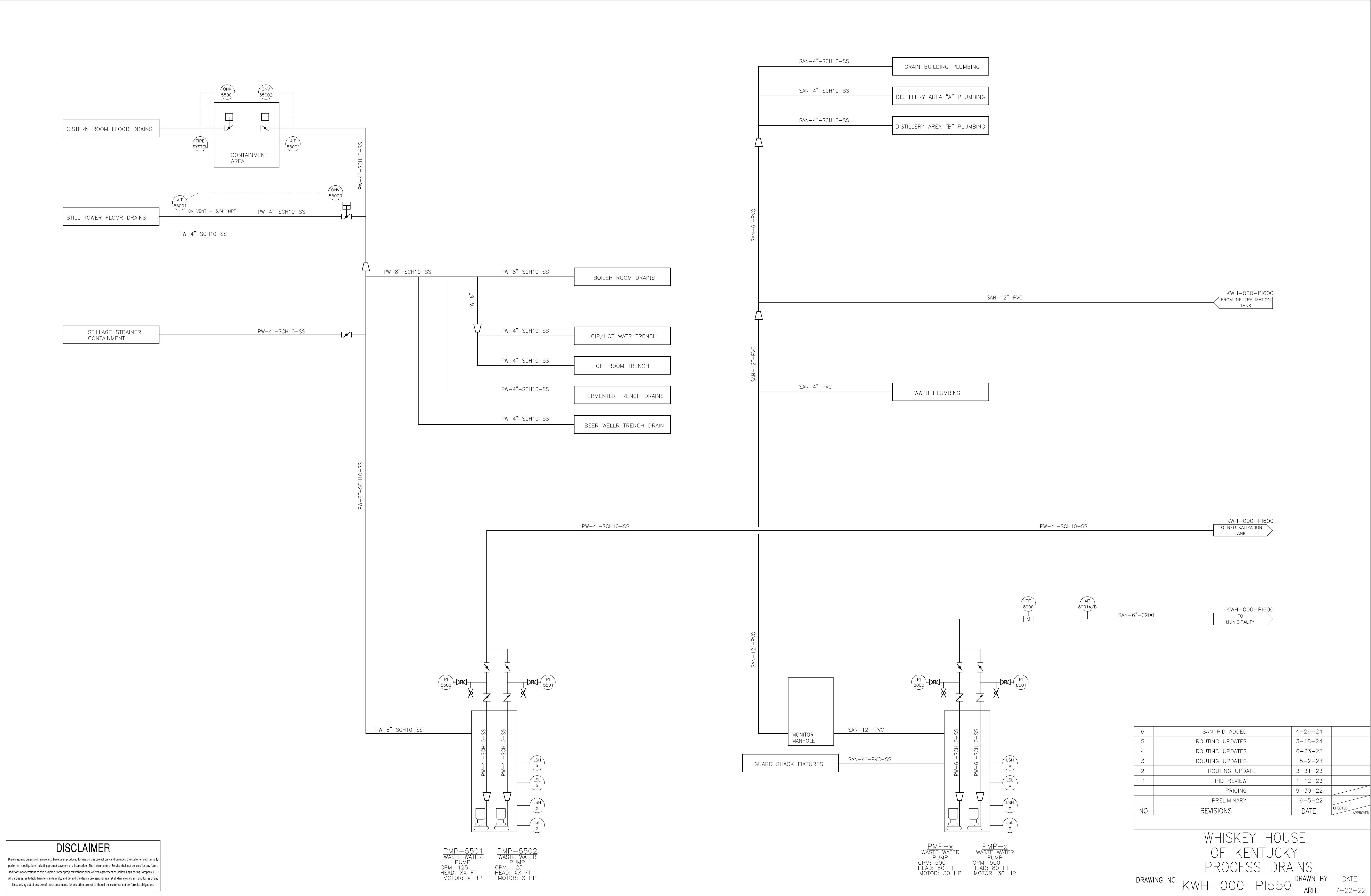
7	ROUTING UPDATES	11-14-23	
5	ROUTING UPDATES	7-28-23	
4	ROUTING UPDATES	6-23-23	
3	ROUTING UPDATES	5-2-23	
2	ROUTING UPDATE	3-31-23	
1	PID REVIEW	1-12-23	
	PRICING	9-30-22	
	PRELIMINARY	9-5-22	
NO.	REVISIONS	DATE	CHECKED APPROVED

DISCLAIMER

Drawings, instruments of service, etc. have been produced for use on this project only and provided the customer substantially performs its obligations including prompt payment of all sums due. The Instruments of Service shall not be used for any future additions or alterations to this project or other projects without prior written agreement of Harlow Engineering Company, LLC. All parties agree to hold harmless, indemnify, and defend the design professional against all damages, claims, and losses of any kind, arising out of any use of these documents for any other project or should the customer not perform its obligations.

WHISKEY HOUSE
OF KENTUCKY
COMPRESSED AIR

DRAWING NO. KWH-000-PI540 DRAWN BY ARH DATE 7-22-22



Kentucky Whiskey House
160 GPM RO SYSTEM
& QUAD CARBON FILTER
& TRIPLEX HCE SOFTENERS

CULLIGAN INTERNATIONAL
PROJECT#: RL0008

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Let.	Change	By	App	Date		DETAILED BY:		APP. BY:	SHEET
A	SUBMITTED FOR APPROVAL	MDG	DKR	04/17/23		MDG 04/17/2023	DKR 04/17/2023		1 OF 5
						REF. NO.		DRAWING NO.	
						RL0008		RL0008-PID-001	

	CARTRIDGE FILTER FILTERITE TYPE		PUMP METERING
	COMPRESSOR CENTRIFUGAL		PUMP PISTON
	COMPRESSOR PISTON AIR		PUMP CENTRIFUGAL
	DEGASIFIER DECARBONATOR		PUMP SUMP
	ELECTRIC MOTOR DRIVE		MOTOR HORIZONTAL
	ENERGY RECOVERY TURBINE		MOTOR VERTICAL
	EDUCTOR EJECTOR		MIXER AGITATOR
	FILTER BOWL TYPE		MIXER STATIC
	FILTER STRAINER		PRESSURE VESSEL
	HX PLATE AND FRAME		TANK COVERED BRINE
	HX SHELL AND TUBE		TANK DOMED STORAGE
	UV STERILIZER		TANK PORTABLE ION EXCHANGE
	PULSATION DAMPENER BOTTLE		TANK NOZZLE
	PULSATION DAMPENER INLINE		
	PRESSURE TAP		
	REVERSE OSMOSIS		
	INJECTION QUILL		

	ANGLE		THREE WAY
	BALL CLOSED		SPRING CHECK
	BALL OPEN		
	BUTTERFLY CLOSED		
	BUTTERFLY OPEN		
	CHECK		
	COCK		
	DIAPHRAGM CLOSED		
	DIAPHRAGM OPEN		
	FOUR WAY		
	GATE CLOSED		
	GATE OPEN		
	GLOBE CLOSED		
	GLOBE OPEN		
	KNIFE GATE		
	NEEDLE		
	PLUG CLOSED		
	PLUG OPEN		
	STRAIGHT THROUGH PRESSURE RELIEF		
	PRESSURE SAFETY RELIEF		
	VACUUM SAFETY RELIEF		
	REDUCER		
	RESTRICTION ORIFICE		

	CYLINDER PISTON WITH LIMIT STOP ON OPEN
	DIAPH P BAL W/LIMIT STOP ON OPEN
	DIAPH P BAL AIR CLOSE AIR OPEN
	DIAPH P BAL AIR CLOSE LINE P OPEN
	DIAPH P BAL AIR CLOSE SPRING OPEN
	DIAPH P BAL AIR OPEN LINE P CLOSE
	DIAPH P BAL AIR OPEN SPRING CLOSE
	DIAPHRAGM PRESSURE BALANCED
	DIAPHRAGM WITH PNEUMATIC POSIT
	DIAPHRAGM WITH POSITIONER
	DIAPHRAGM
	DOUBLE ACTING W/O POSITIONER
	ELECTRO HYDRAULIC
	HAND ACTUATOR OR HANDWHEEL
	HYDRAULIC OPERATED
	REGULATOR BACKPRESSURE EXTERNAL
	REGULATOR BACKPRESSURE INTERNAL
	REGULATOR P REDUCING EXTERNAL
	REGULATOR P REDUCING INTERNAL
	REGULATOR TEMPERATURE
	SINGLE ACTING WITHOUT POSITIONER
	SOLENOID OPERATED
	SPRING ACTUATED RELIEF (SAFETY)
	MOTOR OPERATED

	ALARM FIELD MOUNTED
	ALARM MAIN MOUNTED
	INTERLOCK
	LOCAL MOUNTED
	PANEL MOUNTED
	PILOT LIGHT FIELD ACCESS
	PILOT LIGHT MAIN ACCESS
	PLC FIELD ACCESSIBLE
	PLC MAIN ACCESSIBLE
	ROTAMETER
	TURBINE FLOW METER
	SONIC FLOW METER

	CONNECTION TO PROCESS —OR— INSTRUMENT SUPPLY
	VARYING PNEUMATIC SIGNAL
	ELECTRICAL SIGNAL
	CAPILLARY TUBING (FILLED SYSTEM)
	HYDRAULIC SIGNAL
	TUBING
	SKID LIMIT
	"PUNE" PRIMARY PIPING ("BY CULLIGAN")
	HEAT TRACE
	PIPING ("BY OTHERS")

XX-XXXX-XX-XXXX-XX
 LINE SIZE
 PIPE CLASS
 LINE SERVICE
 LINE #
 INSULATION
PIPE CLASS
 ABG - CRYOGENIC (304SS)
 ABN - SWEET PROCESS (CARBON STEEL)
 ABX - CORROSIVE FLUIDS (316SS)
 AD4H - CHLORINE FEED (HASTALOY C)
 AF - CORROSIVE FLUIDS (FIBERGLASS)
 AJ - DOMESTIC WATER (GALVANIZED)
 AST3 - BFW (CARBON STEEL)
 AJUN - CATEGORY D UTILITIES (CARBON STEEL)
 BS3 - BOILER FEED (CARBON STEEL)
 BS4 - BOILER FEED (CARBON STEEL)

BF - BOILER FEED
CD - CLOSED DRAIN
CH - CHEMICALS
FR - FRESH WATER
FG - FUEL GAS
G - GAS (NATURAL)
IA - INSTRUMENT AIR
K - CAUSTIC
PR - PRESSURE RELIEF
WA - POTABLE WATER

ABF	—	ACTUATED BUTTERFLY VALVE	KV	—	KNIFE [GATE] VALVE
AE	—	ANALYZER ELEMENT	LC	—	LEVEL CONTROL
AI	—	ANALOG INPUT	LS	—	LEVEL SWITCH
AIT	—	ANALYSIS INDICATOR TRANSMITTER	LSL	—	LEVEL SWITCH LOW
AO	—	ANALOG OUTPUT	LSH	—	LEVEL SWITCH HIGH
AR	—	ANALYSIS RECORDER	LT	—	LEVEL TRANSMITTER
AT	—	ANALYZER TRANSMITTER	M	—	MOTOR
ATM	—	ATMOSPHERE	MCC	—	MOTOR CONTROL CENTER
AV	—	ANGLE VALVE	MCP	—	MASTER CONTROL PANEL
BFP	—	BACKFLOW PREVENTER	MH	—	MEMBRANE HOUSING
BFV	—	BUTTERFLY VALVE	MMF	—	MULTIMEDIA FILTER
BV	—	BALL VALVE	MP	—	METERING PUMP
CE	—	CONDUCTIVITY ELEMENT	NV	—	NEEDLE VALVE
CI	—	CONDUCTIVITY INSTRUMENT	P	—	PUMP OR COMPRESSOR
CL	—	CLEAN IN PLACE	PCV	—	PRESSURE REDUCING OR REGULATING VALVE W/ SWITCH
CMP	—	CHEMICAL METERING PUMP	PD	—	PRESSURE DIFFERENTIAL INDICATOR
COND	—	CONDUCTIVITY	PDS	—	PRESSURE DIFFERENTIAL SWITCH
CP	—	CENTRIFUGAL PUMP	PDT	—	PRESSURE DIFFERENTIAL TRANSMITTER
CR	—	CONDUCTIVITY RECORDER	PE	—	PRESSURE ELEMENT
CS	—	CONDUCTIVITY SWITCH	PF	—	PERMEATE FLUSH
CV	—	CHECK VALVE	pHE	—	pH ELEMENT
DI	—	DIGITAL INPUT	pHl	—	pH INDICATOR
DO	—	DIGITAL OUTPUT	PI	—	PRESSURE INDICATOR
DV	—	DIAPHRAGM VALVE	PV	—	PILOT VALVE
EH	—	ELEMENT HOUSING	PLC	—	PROGRAMMABLE LOGIC CONTROLLER
EV	—	ELECTRICALLY ACTUATED VALVE	POC	—	POINT-OF—CONTACT
F	—	FILTER OR CARTRIDGE STRAINER	PR	—	PRESSURE REGULATOR
FC	—	FAIL CLOSED	PRV	—	PRESSURE RELIEF VALVE
FCV	—	FLOW CONTROL VALVE	PS	—	PRESSURE SWITCH
FE	—	FLOW ELEMENT	PSH	—	PRESSURE SWITCH HIGH
FG	—	FLOW SIGHT GLASS	PSL	—	PRESSURE SWITCH LOW
FI	—	FLOW INDICATOR	PSV	—	PRESSURE SAFETY VALVE
FO	—	FLOW ORIFICE	PT	—	PRESSURE TRANSMITTER
FQI	—	FLOW TOTALIZING INDICATOR	PU	—	PLUG VALVE
FQIS	—	FLOW TOTALIZING INDICATOR W/ SWITCH	RO	—	RESTRICTION ORIFICE
FR	—	FLOW RECORDER	SB	—	SOLENOID BLOCK
FS	—	FLOW SWITCH	SM	—	STATIC MIXER
FT	—	FLOW TRANSMITTER	ST	—	STEAM TRAP
FV	—	FOOT VALVE	SOV	—	SOLENOID VALVE
GLV	—	GLOBE VALVE	SV	—	SAMPLE VALVE
GV	—	GATE VALVE	T	—	TANK
HARD	—	HARDNESS	TCV	—	TEMPERATURE CONTROL VALVE
HCV	—	HAND CONTROL VALVE	TE	—	TEMPERATURE ELEMENT
HDA	—	HAND/OFF/AUTO SWITCH	TI	—	TEMPERATURE INDICATOR
HOF	—	HAND SWITCH ON/OFF	TP	—	TIE-IN POINT
HS	—	HAND SWITCH	TURB	—	TURBIDITY
HTR	—	HEATER	V	—	VALVE
IA	—	INSTRUMENT AIR	VFD	—	VARIABLE FREQUENCY DRIVE
IL	—	INDICATING LIGHT	XV	—	SOLENOID VALVE
IQ	—	INJECTION QUILL	ZC	—	NORMALLY CLOSED
			ZO	—	NORMALLY OPEN

DO NOT SCALE DRAWING
TOLERANCES: $\pm 1/8"$ UNLESS OTHERWISE NOTED

Let.	Change	By	App	Date
A	SUBMITTED FOR APPROVAL	MDG	DKR	04/17/23

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ENGINEERED SYSTEMS
LIBERTYVILLE, ILLINOIS

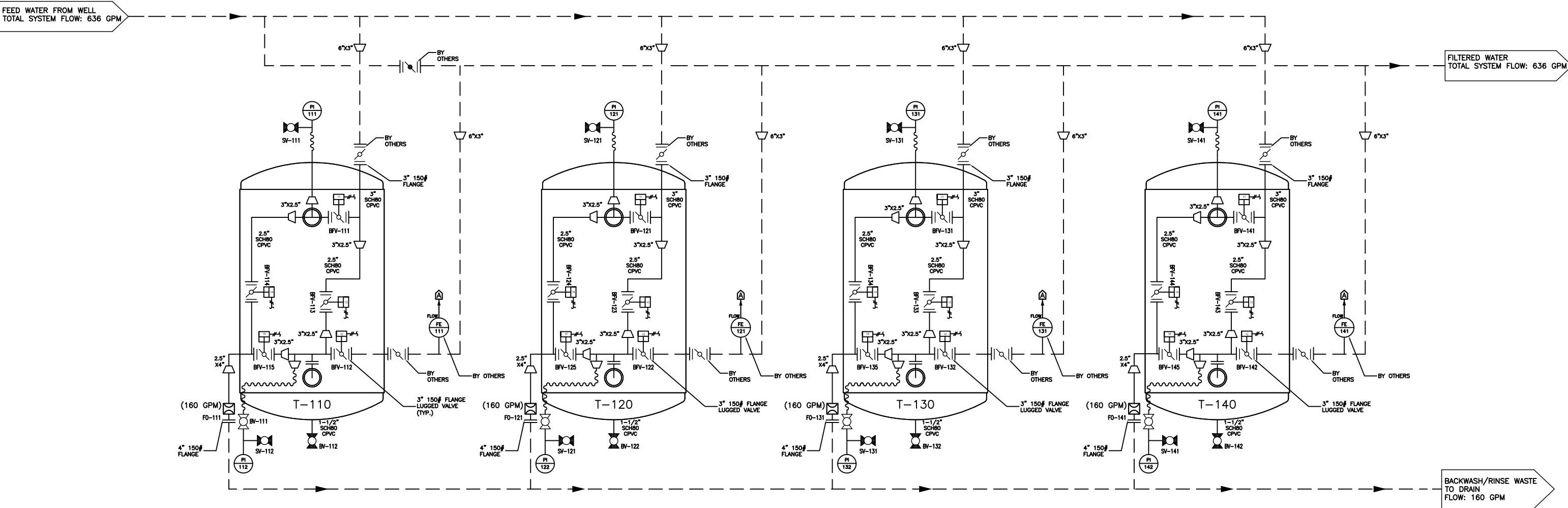
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NAME	PIPING & INSTRUMENTATION DIAGRAM Kentucky Whiskey House
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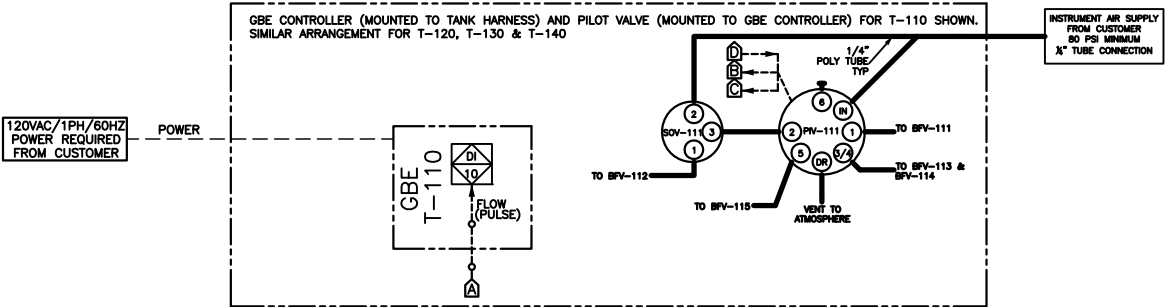
NOMENCLATURE PAGE

DETAILED BY: MDG 04/17/2023	APP. BY: DKR 04/17/2023	SHEET 2 OF 5
REF. NO. RL0008	DRAWING NO. RL0008-PID-001	

DASHED LINE PIPING IS BY
OTHERS



QUAD PROGRESSIVE CARBON FILTERS
CULLIGAN HR-543-NC



VALVE SEQUENCE CHART CARBON FILTERS			T-110, T-120, T-130 & T-140				
			BFV-111/ BFV-121/ BFV-131/ BFV-141	BFV-112/ BFV-122/ BFV-132/ BFV-142	BFV-113/ BFV-123/ BFV-133/ BFV-143	BFV-114/ BFV-124/ BFV-134/ BFV-144	BFV-115/ BFV-125/ BFV-135/ BFV-145
CYCLE	FLOW RATE	DURATION	VALVE 1	VALVE 2	VALVE 3	VALVE 4	VALVE 5
SERVICE	A/R	A/R	0	0	X	X	X
BACKWASH	160 GPM	10 MIN *	X	X	0	0	X
SETTLE	0 GPM	2 MIN	X	X	X	X	0
FAST RINSE	160 GPM	10 MIN *	0	X	X	X	0
STANDBY	0 GPM	A/R	0	X	X	X	X

O = OPEN X = CLOSED

* THESE VALUES
ARE THEORETICAL
AND WILL VARY
BASED ON SPECIFIC
SITE CONDITIONS
INCLUDING INCOMING
PRESSURE, FLOW
RATE, ETC...

T-110 THRU T-140
MATERIAL: CARBON STEEL
EPOXY COATED
100 PSI NON-CODE TANKS
SIDE INLET, BOTTOM OUTLET
54" O.D. X 60" SIDESHELL
INTERNAL LINING: DEVOE 233H
EXTERIOR COATING: TNEDEC N69F
EPOXY COATING

NOTES:
1. INTERCONNECTING PIPING AND MANUAL ISOLATION VALVES BY OTHERS (SHOWN IN DASHED LINES)

DO NOT SCALE DRAWING TOLERANCES: ±1/8" UNLESS OTHERWISE NOTED			
Let.	Change	By	App Date
A	SUBMITTED FOR APPROVAL	MDG	DKR 04/17/23

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NAME PIPING & INSTRUMENTATION DIAGRAM Kentucky Whiskey House QUAD CARBON FILTERS		
DETAILED BY: MDG 04/17/2023	APP. BY: DKR 04/17/2023	SHEET 3 OF 5
REF. NO. RL0008	DRAWING NO. RL0008-PID-001	

FEED WATER: 645 GPM CONTINUOUS
840 GPM PEAK

INLET CONNECTION
TYPE: FLANGE
SIZE: 3"
RATING: 150#
MATERIAL: PVC, SCH80

OUTLET FILTERED WATER
TO RO SYSTEM
SOFTENED WATER: 645 GPM CONTINUOUS
840 GPM PEAK
(SEE SHEET 5)

OUTLET CONNECTION
TYPE: FLANGE
SIZE: 3"
RATING: 150#
MATERIAL: PVC, SCH80

120VAC/60HZ/1PH
POWER FROM
CUSTOMER

GBE-211

PI
211

SV-211

3900 VALVE

PI
212

SV-212

T-210

1" GREY
HOSE RELIEF
LINE

1/2" PE
HOSE

GBE-221

PI
221

SV-221

3900 VALVE

PI
222

SV-222

T-220

1" GREY
HOSE RELIEF
LINE

1/2" PE
HOSE

GBE-231

PI
231

SV-231

3900 VALVE

PI
232

SV-232

T-230

1" GREY
HOSE RELIEF
LINE

1/2" PE
HOSE

BRINE
TANK
(BY OTHERS)

DRAIN FLOW:
45 GPM
INTERMITTENT DURING
SOFTENER REGENERATION

TRIPLEX SOFTENERS
HCE-1200-3

INLET CONNECTION
TYPE: FLANGE
SIZE: 2"
RATING: 150#
MATERIAL: PVC, SCH80

T-210 THRU T-230
MAKE: CULLIGAN
MODEL: HCE-1200-3
TYPE: TRIPLEX SOFTENER SYSTEM
FLOW RATE: 215 GPM PER VESSEL CONTINUOUS
280 GPM PER VESSEL PEAK
TANK SIZE: 48" X 72"
MATERIAL: FRP
MAXIMUM PRESSURE: 100 PSI

NOTES:

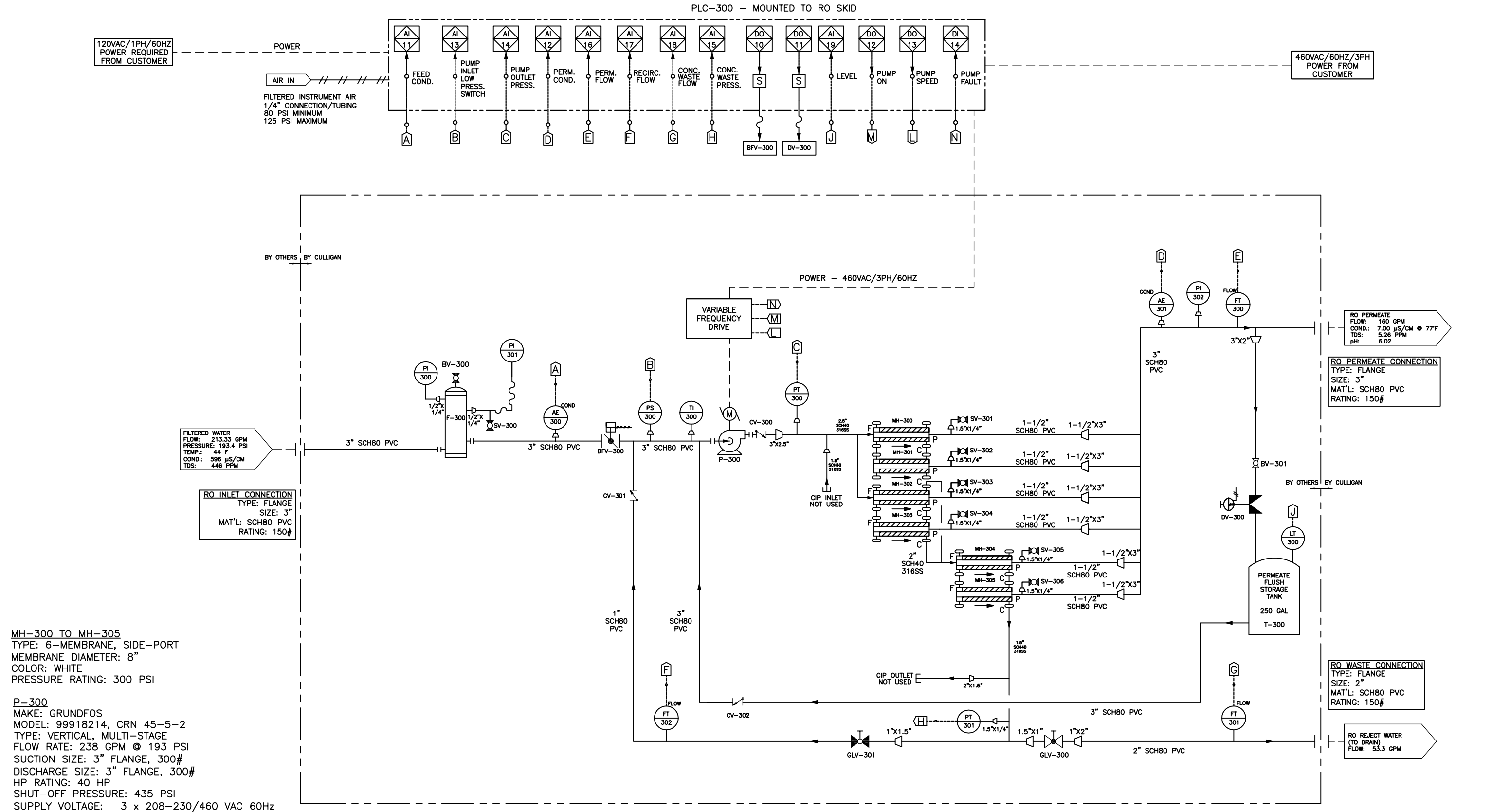
- INTERCONNECTING PIPING, BRINE TANK & VALES SHOWN IN DASHED LINES IS BY OTHERS.
- THE GBE CONTROLLER IS INSTALLED ON REMOTELY & CONNECTED TO EACH OTHER W/ COMMUNICATION CABLE, SEE MANUAL FOR FURTHER DETAILS

DO NOT SCALE DRAWING TOLERANCES: ±1/8" UNLESS OTHERWISE NOTED				
Let.	Change	By	App	Date
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NAME PIPING & INSTRUMENTATION DIAGRAM Kentucky Whiskey House HCE-1200-3 TRIPLEX SOFTENER SYSTEM		
DETAILED BY: MDG 04/17/2023	APP. BY: DKR 04/17/2023	SHEET 4 OF 5
REF. NO. RL0008	DRAWING NO. RL0008-PID-001	



MH-300 TO MH-305
TYPE: 6-MEMBRANE, SIDE-PORT
MEMBRANE DIAMETER: 8"
COLOR: WHITE
PRESSURE RATING: 300 PSI

P-300
MAKE: GRUNDFOS
MODEL: 99918214, CRN 45-5-2
TYPE: VERTICAL, MULTI-STAGE
FLOW RATE: 238 GPM @ 193 PSI
SUCTION SIZE: 3" FLANGE, 300#
DISCHARGE SIZE: 3" FLANGE, 300#
HP RATING: 40 HP
SHUT-OFF PRESSURE: 435 PSI
SUPPLY VOLTAGE: 3 x 208-230/460 VAC 60Hz

F-300
MAKE: SHELCO
MODEL: 22FOS3-3F
CARTRIDGE LENGTH: 30"
CARTRIDGE QUANTITY: 22
RATING: 5 MICRON
MATERIAL: 304SS
MAX PRESSURE: 150 PSI
CONNECTION SIZE: 3" 150# FLANGE

NOTES:
1. FOR MH-300 THRU MH-305, F=FEED, C=CONCENTRATE, & P=PERMEATE.

DO NOT SCALE DRAWING TOLERANCES: ±1/8" UNLESS OTHERWISE NOTED				
Let.	Change	By	App	Date
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NAME PIPING & INSTRUMENTATION DIAGRAM KENTUCKY WHISKEY HOUSE ELIZABETHTOWN, KY RO SKID ASSEMBLY		
DETAILED BY: MDG 04/17/2023	APP. BY: DKR 04/17/2023	SHEET 5 OF 5
REF. NO. RL0008	DRAWING NO. RL0008-PID-001	